

* PSU Question (15 Question)

~~problem~~

```
1. void fun(int x)
{
    x = x + 10;
}
```

```
int main()
{
    int a = 5;
    fun(a);
    printf("%d", a);
}
```

Output = ?

A) 5 B) 10 C) 15 D) variable

2. #include <stdio.h>

```
int fun(int x)
{
    return x * x;
}
```

```
int main()
{
    printf("%d", fun(fun(2)));
}
```

output ?

A) 4 B) 8 C) 16 D) 32

3. Which statement about actual and formal parameter is correct?

Ans. Actual parameters are passed during call.

4. Where are local variable stored?
Ans. (a) Heap (b) Stack (c) Data Segment (d) Code Segment

5. Which memory area is used for function call?

(a) Heap (b) Stack (c) Data Segment (d) Register

6. Which memory segment grows and shrinks automatically during program execution?

(a) Heap (b) Stack (c) Code Segment (d) Data Segment

7. Stack follow which principle?

(a) FIFO (b) LIFO (c) Random Access (d) Priority

(8) Which memory area is used by malloc?

(a) Stack (b) Heap (c) Code Segment (d) Register

9. What is stack overflow?

Ans Stack exceeds its allocated size.

10. Which memory segment stores program instructions?

Ans Code segment (Text segment)

11. static int x = 10; where x is stored?

Ans Data Segment

12. Which memory area is used for dynamic memory allocation?

Ans Heap

13. What happens when a function is called?

Ans A new stack frame is created.

14.

14. Which memory segment contains uninitialized global variable?

Ans BSS segment.

15. Which memory area typically grows downward in memory address?

Ans Register.